CIS 5210
HW 3: Informed Search

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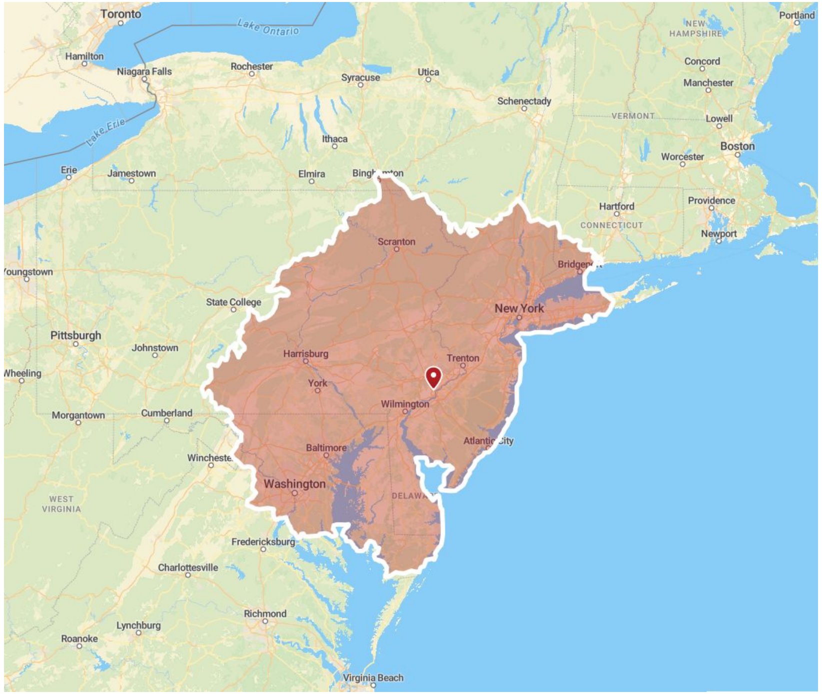
Slides adapted from previous years

Uninformed Search

Breadth-first
Uniform-cost
Depth-first
Depth-limited
Iterative Deepening Depth-first

Philly to DC? =>



3 hour driving isochrome from Philadelphia

Informed Search

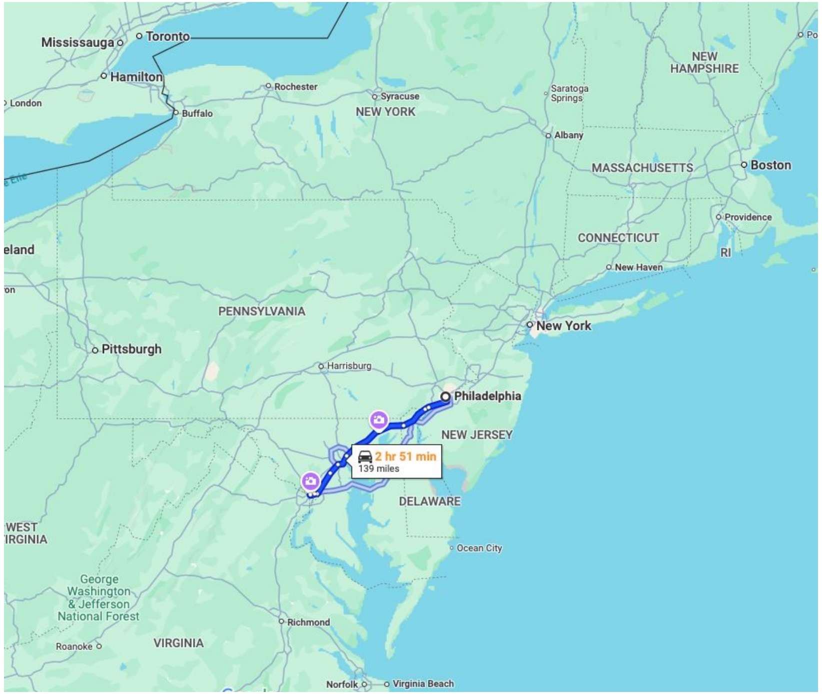
Use heuristic to:

- Refine exploration direction
- Explore more promising paths first
- Reduce computational cost

Greedy Best-first

A*

Philly to DC =>



Google Maps route from Philadelphia to Washington D.C.

Penn Engineering

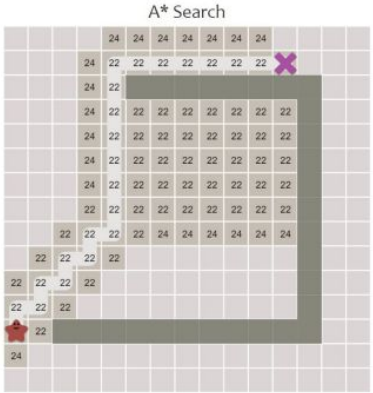
Costs and Heuristics

g(n): actual cost from start to n (path cost)

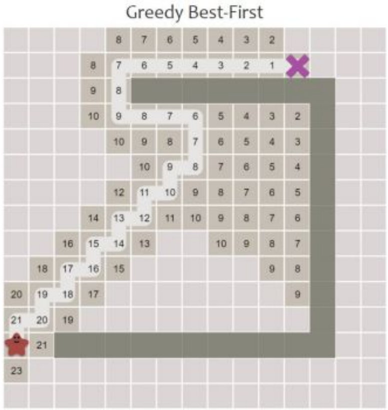
h(n): estimated cost from n to goal (heuristic)

f(n) = g(n) + h(n) (estimated cost from start to goal)

A*: f(n)

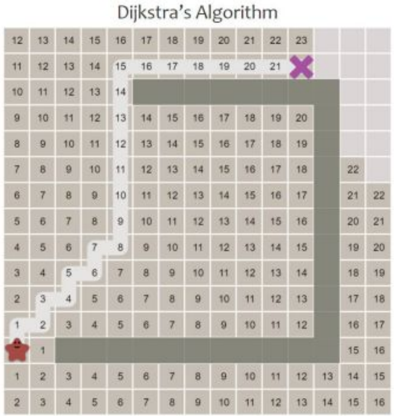


Greedy Best-first: h(n)



not optimal

Uniform Cost: g(n)



optimal but inefficient

Heuristic Admissibility

A* is optimal IF AND ONLY IF h(n) is admissible

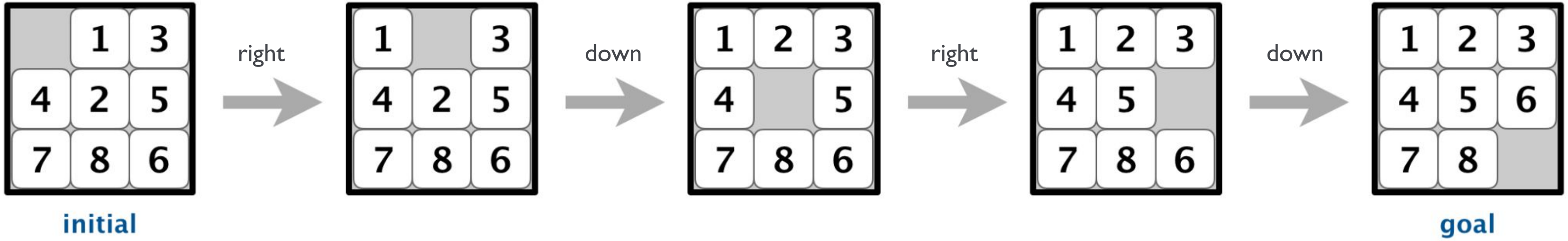
An admissible heuristic never _overestimates_ the cost from n to the goal

Techniques to design heuristics:

- Solution of a relaxed problem
- Exploit knowledge of the domain

(see Ed post #113 for the implications of consistency)

Tile Puzzle



Board representation: list of list of integers, e.g. ([0, 1, 3], [4, 2, 5], [7, 8, 6])

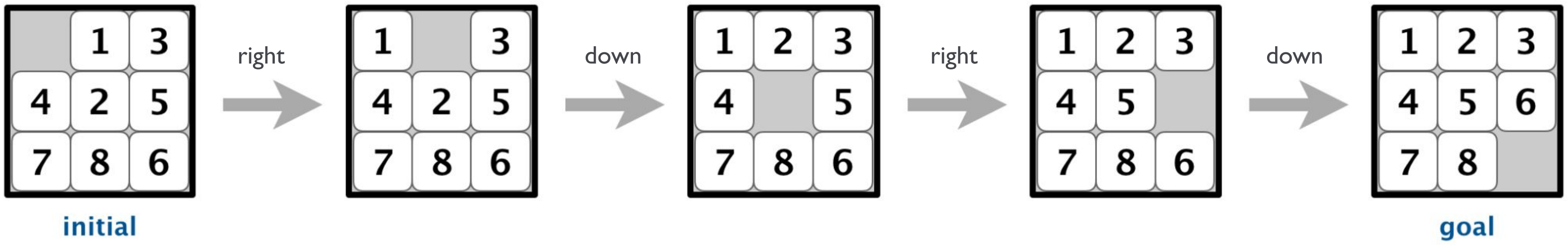
Solved state: numerically ordered, empty space in bottom right

Moves ({up, down, left, right}) swap empty space with tile in given direction

Solutions are sequences of moves to get to solved state

e.g. [right, down, right, down]

Tile Puzzle



First method: IDDFS

- Depth limited DFS subroutine
- Return <u>ALL</u> optimal solutions

Second method: A*

- Heuristic: summed Manhattan distance of all tiles to correct positions
- Return <u>any</u> optimal solution

Grid Navigation

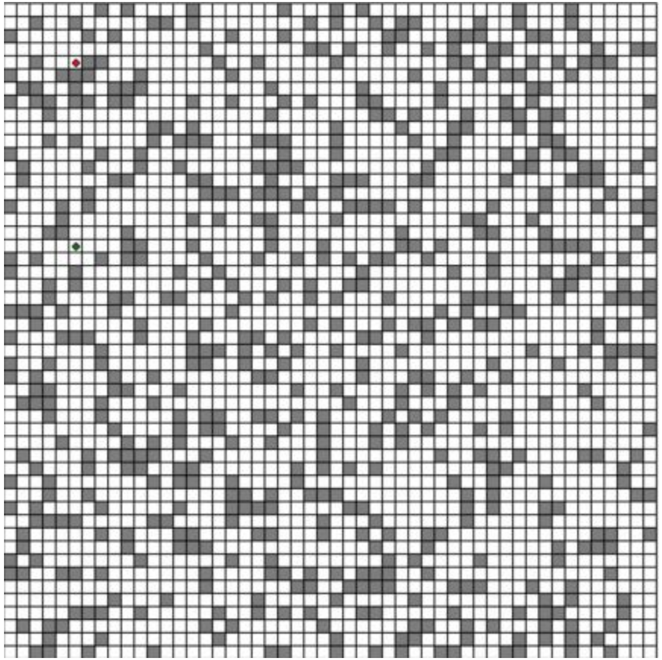
Grid defined by scene file:

empty cells: .

barrier cells: X

scene_simple.txt
```
.......
..XXX..
.XXXXX.
.XXXXX.
.XXXXX.
..XXX..
.......
```

Represent as list of list of booleans, X => True



Moves: vertical, horizontal, diagonal (8 total)

A* with Euclidean heuristic. Why?

Why not Manhattan distance?

Linear Disk Movement

Same setup as Distinct Linear Disk Movement from HW2

Use A* to return the optimal solution (list of moves)

Design a heuristic:

- Must be admissible to ensure optimality
- Must be informative enough to significantly improve performance
- What rule in the problem can be relaxed to get an intuitive h(n)?

Questions?